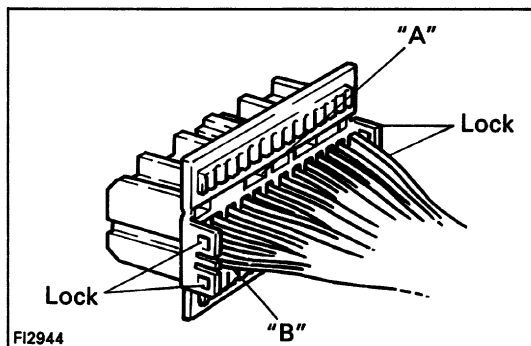


Electronic Control Module (ECM)

INSPECTION OF ECM

HINT: The MR and SFI circuit can be checked by measuring the resistance and voltage at the wiring connectors of the ECM.



1. PREPARATION

- Disconnect the connectors from the ECM.
- Remove the locks as shown in the illustration so that the tester probe(s) can easily come in.

NOTICE: Pay attention to sections "A" and "B" in the illustration which can be easily broken.

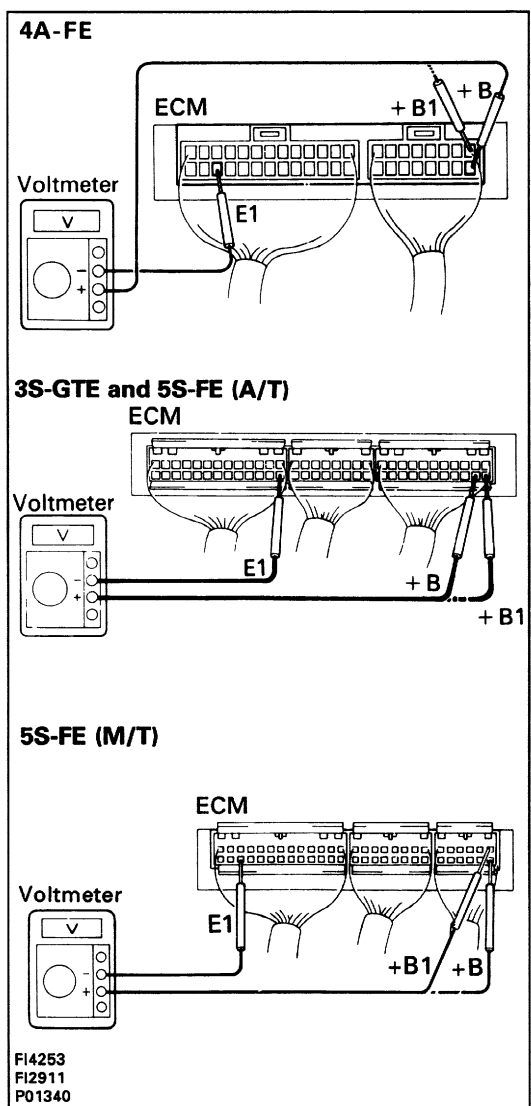
2. INSPECT VOLTAGE OF ECM

Check the voltage between each terminal of the wiring connectors.

- Turn the ignition switch ON.
- Measure the voltage at each terminal.

HINT:

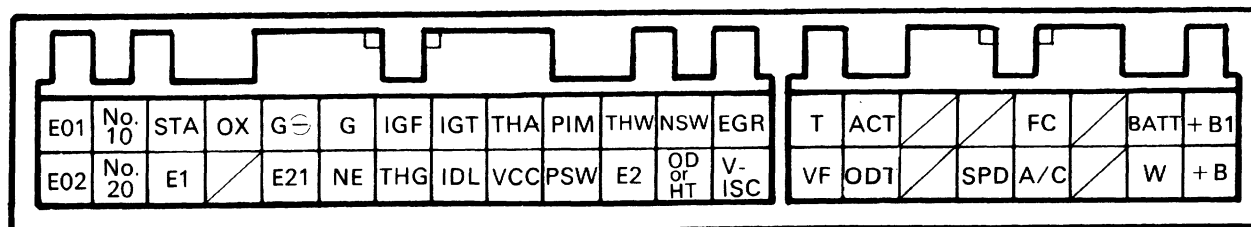
- Perform all voltage measurements with the connectors connected.
- Verify that the battery voltage is 11 V or more when the ignition switch is ON.



Voltage at ECM Wiring Connectors (4A-FE)

Terminals	Condition		STD voltage (V)
+B + B1 – E1	IG SW ON		10–14
BATT – E 1	–		10–14
I D L – E2	IG SW ON	Throttle valve open	10–14
PSW – E2		Throttle valve fully closed	10–14
PIM–E2	IG SW ON		3.3–3.9
VCC – E2			4.5–5.5
No.10 E01 No.20 E02			10–14
THA – E2	IG SW ON	Intake air temp. 20°C (68°F)	1 –3
THW – E2		Engine coolant temp. 80°C (176°F)	0.1 –1.0
STA – E1	Cranking		6–14
IGT – E1	Cranking or idling		0.7–1.0
W – E1	No trouble (Malfunction indicator– lamp off) and engine running		10–14
A/C – E1	IG SW ON	Air conditioning ON	8–14
ACT – E1		Heater blower SW ON	4–6
T – E1		Data link connector 1 TE1 – E1 not connected	10–14
		Data link connector 1 TE1 – E1 connected	0.5 or less
NSW – E1		Park/Neutral position switch P or N position	0 – 2
		Ex. park/neutral position switch P or N position	6–14
V–ISC – E1	Cranking for ten seconds after starting		10–14

ECM Terminals



Voltage at ECM Wiring Connectors (3S-GTE)

Terminals	Condition		STD voltage (V)
+B - E1 + B1	IG SW ON		10-14
BATT - E 1	-		10-14
I D L - E2	IG SW ON	Throttle valve open	4.5-5.5
VTA - E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.1 -1.0
		Throttle valve fully open	3.2-4.2
VC - E2		-	4.5-5.5
VS - E2		Measuring plate fully closed	3.7-4.3
		Measuring plate fully open	0.2-0.5
	Idling	1.6-4.1	
	3,000 rpm	1.0-2.0	
No. 1 No. 2 - E01 No. 3 - E02 No. 4	IG SW ON		10-14
THA - E2	IG SW ON	Intake air temp. 20°C (68°F)	1 -3
THW - E2		Engine coolant temp. 80°C (176°F)	0.1 -1.1
STA - E1	Cranking		6-14
IGT - E1	Cranking or idling		0.8-1.2
RSC - E1 RSO	IG SW ON	ECM connectors disconnected	8-14
W-EI	No trouble (malfunction indicator lamp off) and engine running		10-14
PIM - E2	IG SW ON		2.5-4.5
AC -E1	IG SW ON	Air conditioning ON	8-14
*1 TVIS - E1		Throttle valve fully closed	2.0 or less
		Throttle valve open	10-14
*2 TVIS - E1	Idling		2.0 or less
	4,200 rpm or more		10-14
TE1 - E1	IG SW ON	Data link connector 1 TE1 - E1 not connected	10-14
		Data link connector 1 TE1 - E1 connected	0.5 or less

ECM Terminals

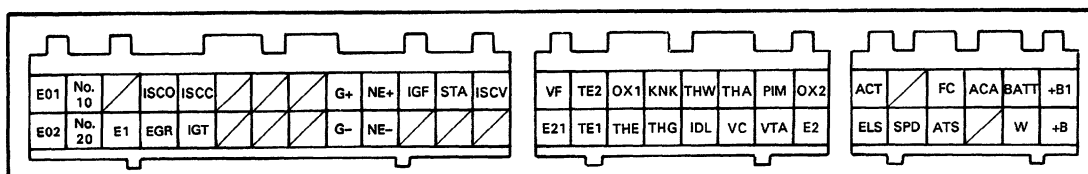
*1 w/ Regular Unleaded Gasoline *2 w/ Premium Unleaded Gasoline

E01	No.1	No.2	RSO	RSC	HT	STJ	EGR	G2	NE	IGF	TPC	TVIS	VF	OX	PIM	THW	THA	VS	VC	STA	AC	SPD	ATS	FPR	W	STP	ELS	BATT
E02	No.3	No.4				IGT		G1	G⊖			E1	TE1	TE2	KNK	IDL	VTA	THG	E2	ACT						FC	+B1	+B

Voltage at ECM Wiring Connectors (5S-FE)

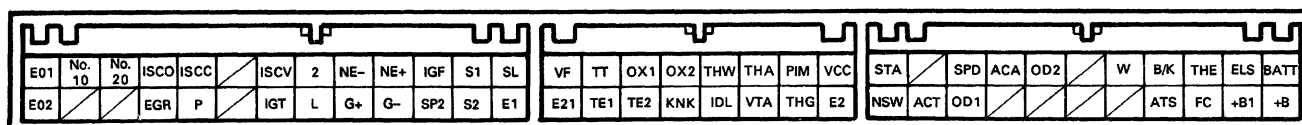
Terminals	Condition		STD voltage (V)
+B + B1 - E1	IG SW ON		10-14
BATT - E1	-		10-14
IDL - E2	IG SW ON	Throttle valve open	8-14
VTA - E2		Throttle valve fully closed (Throttle opener must be cancelled first)	0.8 - 1.2
		Throttle valve fully open	3.2 - 4.2
PIM-E2	IG SW ON		3.3 - 3.9
VC - E2			4.5 - 5.5
No. 10 E01 No. 20 E02			10 - 14
THA - E2	IG SW ON	Intake air temp. 20°C (68°F)	1.9 - 2.9
THW - E2		Engine coolant temp. 80°C (176°F)	0.1 -1.1
STA - E1	Cranking		6 -14
IGT - E1	Cranking or idling		0.8 - 1.2
ISCC - E1 ISCO	IGSWON	ECM connectors disconnected	8 -14
W-EI	No trouble (malfunction indicator lamp off) and engine running		10 - 14
A/C - E1	IG SW ON	Air conditioning ON	8 -14
ACT - E1		Air conditioning ON	4.5 - 5.5
ACA - E1		Air conditioning ON	2 or less
TE1 - E1		Data link connector 1 TE1 - E1 not connected	10-14
		Data link connector 1 TE1 - E1 connected	1 or less
NSW - E1		Park/Neutral position switch P or N position	0 - 2
		Ex. park/neutral position switch P or N position	6 -14
B/ K - E1	Stop light SW ON (Brake pedal depressed)		10 -14

ECM Terminals

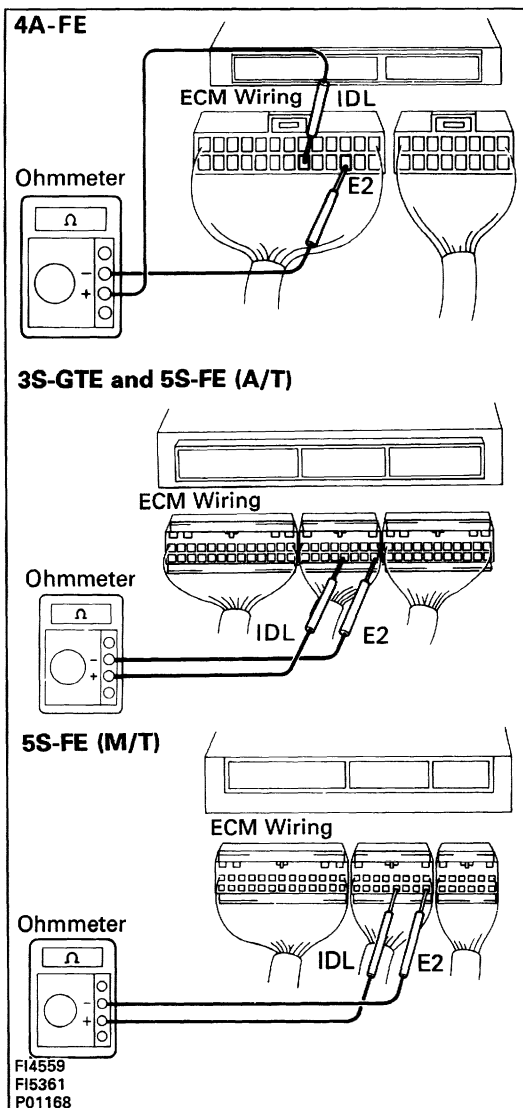


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ECM Terminals



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2. INSPECT RESISTANCE OF ECM

NOTICE:

- Do not touch the ECM terminals.
- The tester probe should be inserted into the wiring connector from the wiring side.

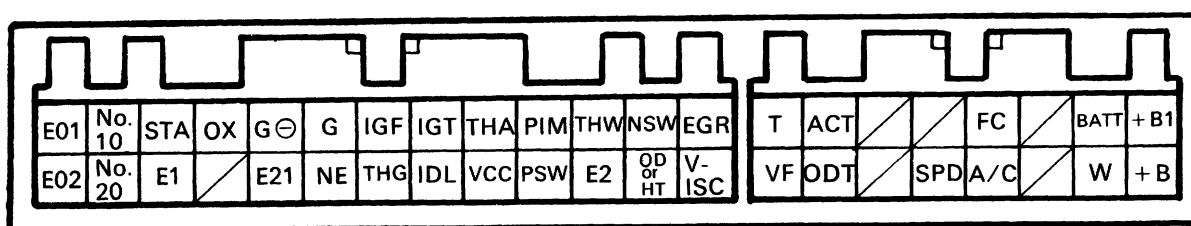
Check the resistance between each terminal of the wiring connectors.

- Disconnect the connectors from the ECM.
- Measure the resistance at each terminal.

Resistance of ECM Wiring Connector (4A-FE)

Terminals	Condition	STD resistance (Ω)
IDL - E2	Throttle valve open	Infinity
	Throttle valve fully closed	0
PSW - E2	Throttle valve fully open	0
	Throttle valve fully closed	Infinity
THA - E2	Intake air temperature 20°C (68°F)	2,000 - 3,000
THW - E2	Engine coolant temperature 80°C (176°F)	200-400
G1 NE - G (-)	Cold	185-265

ECM Terminals

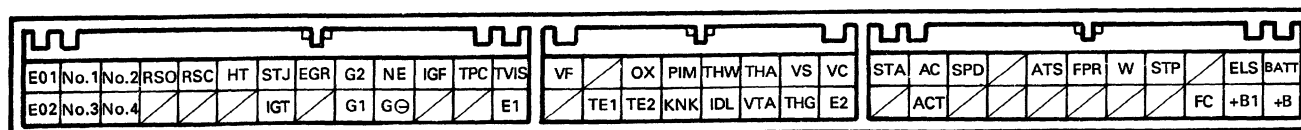


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Resistance of ECM Wiring Connectors (3S-GTE)

Terminals	Condition	STD resistance (Ω)
IDL - E2	Throttle valve open	Infinity
	Throttle valve fully closed	2,300 or less
VTA - E2	Throttle valve fully open	3,100 - 12,100
	Throttle valve fully closed	470-6,100
VC -E2	-	3,900 - 9,000
VS - E2	Measuring plate fully closed	200-600
	Measuring plate fully open	20-1,200
THA - E2	Intake air temp. 20°C (68°F)	2,000 - 3,000
THW - E2	Engine coolant temp. 80°C (176°F)	200-400
G1 G2 - G(-)	Cold	125-190
NE -G(-)	Cold	155-240
RSC + B RSO + B1	-	19.3-22.3

ECM Terminals

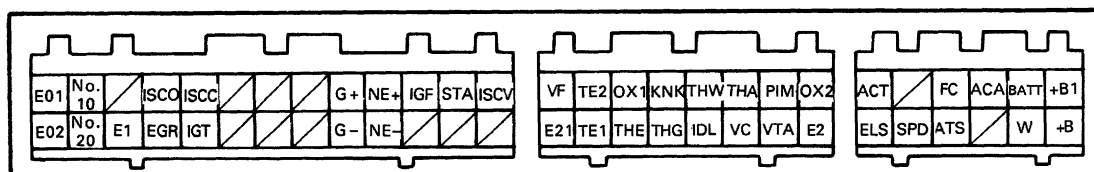


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Resistance of ECM Wiring Connectors (5S-FE)

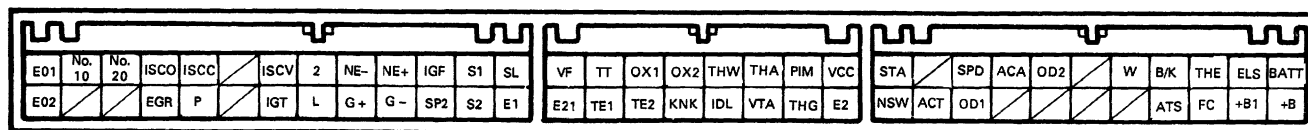
Terminals	Condition	STD resistance (Ω)
IDL - E2	Throttle valve open	Infinity
	Throttle valve fully closed (Throttle opener must be cancelled first)	2,300 or less
VTA - E2	Throttle valve fully open	2,000 - 10,200
	Throttle valve fully closed (Throttle opener must be cancelled first)	200-5,700
VC - E2	—	2,500 - 5,900
THA - E2	Intake air temp. 20°C (68°F)	2,000 - 3,000
THW - E2	Engine coolant temp. 80°C (176°F)	200-400
G+ - G-	Cold	185-265
NE+ - NE-	Cold	370-530
ISCC - + B ISCO - +B I	—	19.3-22.3

ECM Terminals



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ECM Terminals



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